

When using rule of thumb to check if two variances are the same, approximate the significance level, when $n_1=4$, $n_2=3$.

$$\frac{\text{Large } s^2}{\text{Small } s^2} \leq 3 \rightarrow \text{Variances are same}$$

$$\alpha = P(F_{3,2} > 3) + P(F_{2,3} > 3)$$

$$= 0.26 + 0.19 = 0.45$$